Q1 – Survivors vs age, by gender (`q1_survival_vs_age_by_gender.png`)

- Survivors peak in the 20–39 age groups, because most passengers were that age.
- Women outnumber men among survivors in every age group except 0–9, where the counts are equal (19 each).
- Survival rates are about 60–90% for women at most ages, and only about 12–21% for men over 10.
- Boys under 10 are the exception: about 59% survived.
- 177 passengers have no recorded age and are left out of this chart.

Q2 – Fare vs class, with gender and age (`q2_fare_vs_class_gender_age.png`)

- Fare is on a log scale because first-class fares reach about $512.
- Colour shows gender and dot size shows age. The black bars mark the median fare for each gender in each class.
- Median fares drop sharply by class: about $52 (men) and $83 (women) in 1st, $13 and $22 in 2nd, and $8 and $13 in 3rd.
- Women paid more than men in every class.
- The chart leaves out passengers with no age and the 15 passengers with a $0 fare, which a log scale can't show.

Q3 – Survival rate by gender and class (`q3_survival_rate_by_sex_and_class.png`)

Class	Women	Men
1st	97%	37%
2nd	92%	16%
3rd	50%	14%

Women survived far more often than men in every class. Class mattered a lot within each gender: third-class women survived about half as often as first-class women.

Q4 – Who was more likely to survive? (`q4_survival_factors.png`)

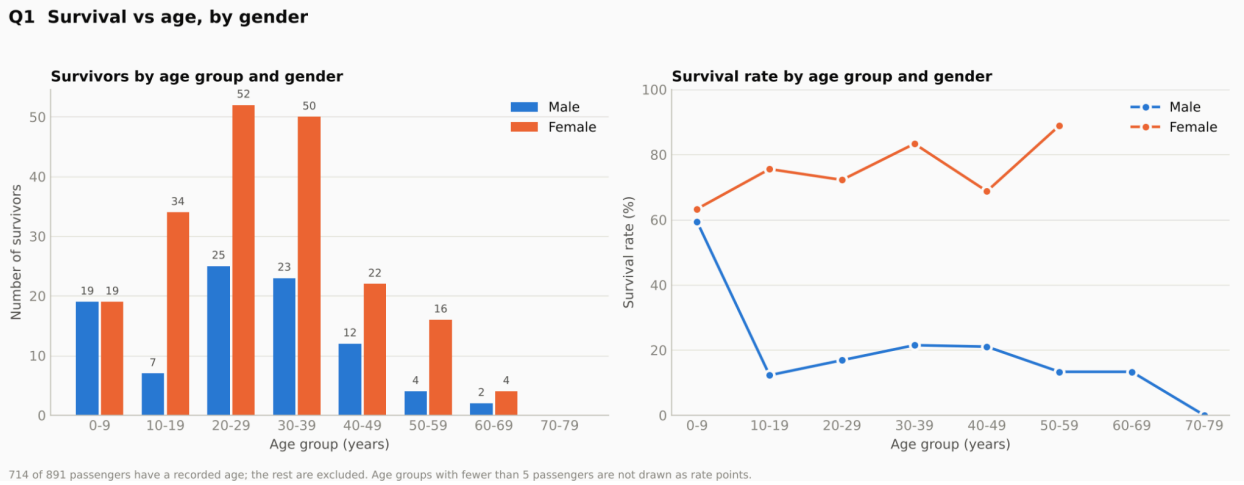
Survival rates by group:

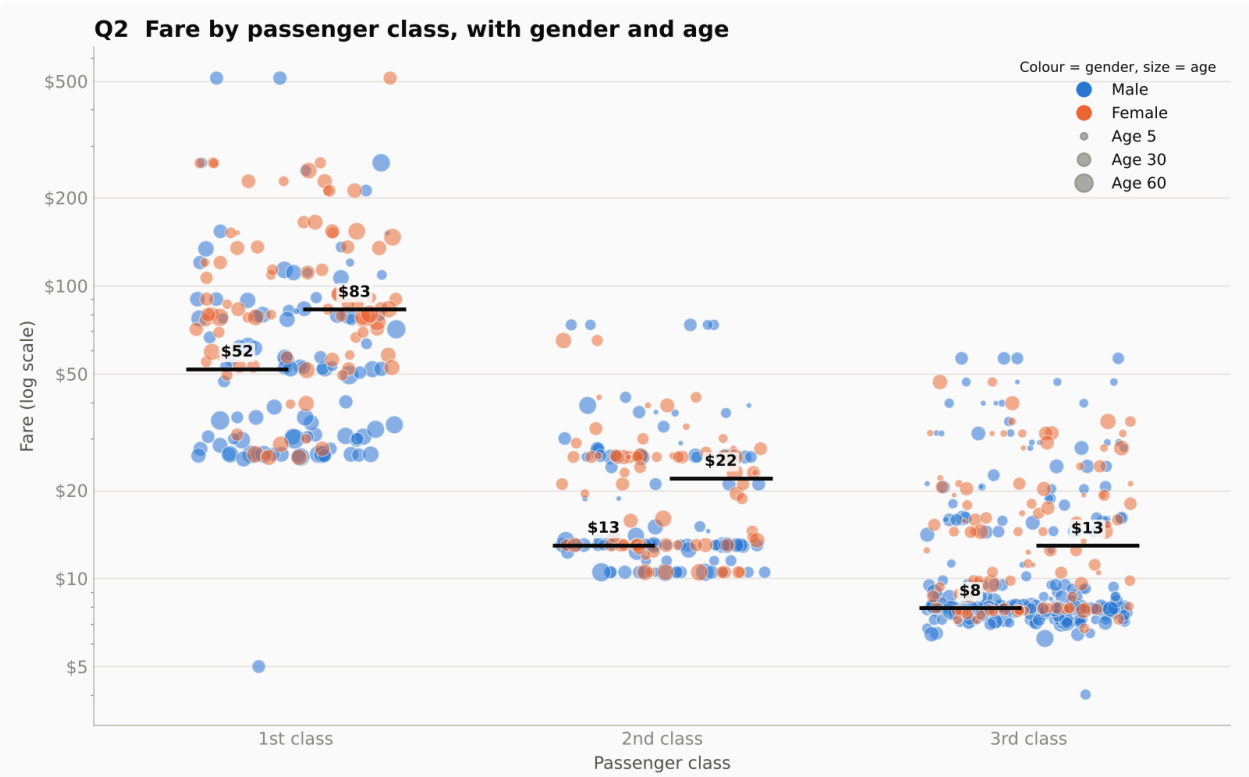
Factor	Higher survival	Lower survival
Gender	Women 74%	Men 19%
Class	1st 63%	3rd 24% (2nd 47%)

Age	Under 13: 58%	60+: 23%
Title in name	Mrs 79%, Miss 70%, Master 57%	Mr 16%
Family size	2–4 people: 58%	Alone 30%, 5+ people 16%
Fare	Priciest quarter 58%	Cheapest quarter 20%

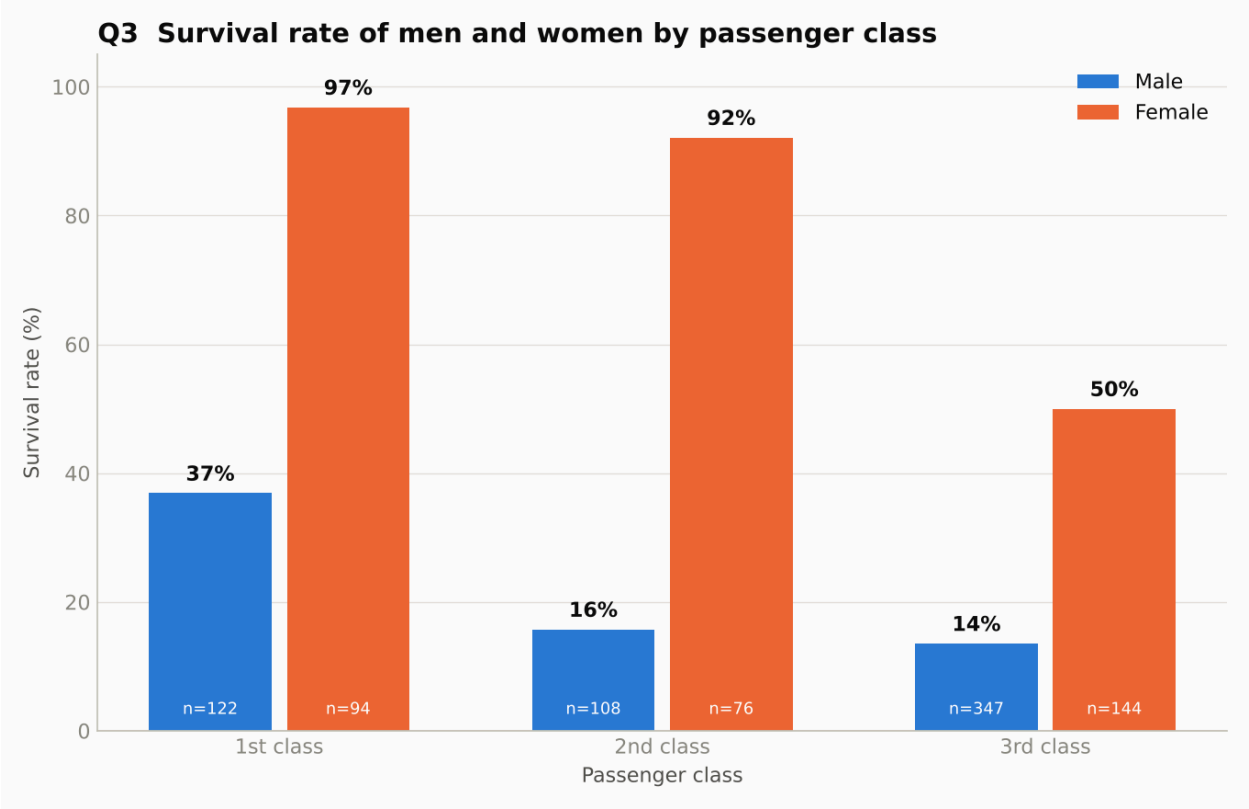
- **Most likely to survive:** women, children, and first-class passengers. A woman in 1st or 2nd class had about a 92–97% chance, and a child in 2nd class survived every time (17 of 17).
- **Least likely to survive:** men, especially third-class men (14%), and people in large families of five or more.
- **Gender and class together:** each effect holds within the other. Women did better in every class, and 1st class did better for both genders.
- **Family size:** it peaks for small groups of 2–4, and travelling alone or in a big family was worse.
- **Fare:** higher fares survived more, but fare mostly tracks class, so it adds little on its own.
- **Caveat:** this shows association, not cause. Class, fare, and cabin location overlap, and that overlap affects how these results should be read. Some groups are also small (only 22 people were 60+, and 23 had other titles).

Visualization part

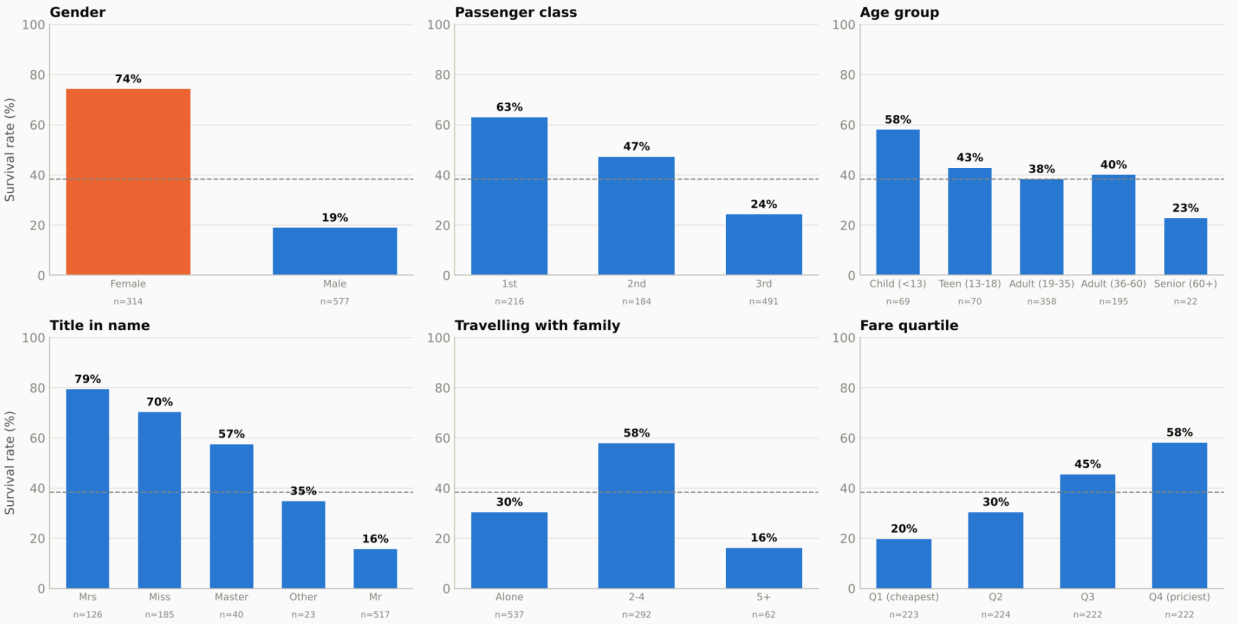




707 passengers shown (known age, fare above $0). Points are jittered sideways within each class. Black bars with labels = median fare (male left, female right).



Q4 Who was more likely to survive?



Dashed line = overall survival rate (38%). n = passengers in each group; age panel excludes passengers with no recorded age.